Paper 2 → Final: Replace-From / Replace-To

Apply these changes to `12th IEEE International Women in Engineering (Paper 2).docx`. **Replace from → replace to.**

Leave everything else untouched: figures (you replace those yourself), the 7 equations, the author block, all 29 references, Tables I, II, III, V, and all captions (`TABLE I.` / `Fig. 1.` come from auto-numbering and stay as they are).

Paragraph and heading changes

Where	Replace from	Replace to
Abstract — last sentence	The practical recommendation is to keep the base model, collect about one-fifth of a field window, and refit the calibrator.	The practical recommendation is to keep the base model and refit its calibrator on about one-fifth of a field window.
Keywords — remove trailing period	Keywords—battery energy storage, calibration, distribution shift, hazard learning, isotonic regression, recalibration.	Keywords—battery energy storage, calibration, distribution shift, hazard learning, isotonic regression, recalibration
Introduction — merge paragraphs 1+2 into one	Laboratory-trained models meet three systematic shifts in the field: partial cycles at varying depth of discharge (DoD), irregular rest intervals, and fluctuating thermal conditions. ¶ ¶ The prognostics literature has mapped state-of-health (SOH) estimation and remaining-useful-life (RUL) prediction in detail, and the calibration-under-shift literature has grown in image and NLP domains. The operational question for batteries has not been answered: when calibration degrades, how much field data restores it?	Laboratory-trained models meet three systematic shifts in the field: partial cycles at varying depth of discharge (DoD), irregular rest intervals, and fluctuating thermal conditions. The prognostics literature has mapped state-of-health (SOH) estimation and remaining-useful-life (RUL) prediction in detail, and the calibration-under-shift literature has grown in image and NLP domains. The operational question for batteries has not been answered: when calibration degrades, how much field data restores it?
Introduction — merge paragraphs 3+4 into one	This paper answers that question with a controlled stress test of a frozen hazard model. Three contributions follow. First, calibration degradation under synthetic operational perturbations is quantified across four horizons and four severities. Second, a recalibration data-requirement curve shows the field-sample fraction needed to restore calibration. ¶ ¶ Third, recalibration is compared against domain-randomized	This paper answers that question with a controlled stress test of a frozen hazard model under the three shifts just described. Three contributions follow. First, we quantify how much calibration degrades under synthetic operational perturbations, spanning four prediction horizons and four severity levels from mild to aggressive. Second, a recalibration data-requirement curve shows the field-sample fraction needed to

retraining, which offers no consistent advantage. The message for operators is a deployment recipe: keep the base model, collect about a fifth of a field window, refit the calibrator.

restore calibration. Third, recalibration is compared against domain-randomized retraining, which offers no consistent advantage over the light recalibration step. For operators this translates to a deployment recipe: refit the calibrator on about a fifth of a field window, keeping the base model frozen.

Related work A — merge two paragraphs into one

Battery health prognostics has concentrated on SOH estimation and RUL prediction. Surveys by Che et al. [1] and Wang et al. [2] catalogue data-driven approaches.

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¶ Representative methods include deep learning for SOH without degradation experiments [3], optimized XGBoost for capacity prediction [4], and cycle-based deep architectures [5]; Massaoudi et al. [6] review deep-learning prognostics. Hybrid and EIS-based RUL models [7]–[9], joint SOH/RUL prediction from single-cycle or discharge-fragment data [10], [11], and transfer learning for fast-charging profiles [12] round out the recent work. A common limitation is that validation is confined to clean laboratory splits of a single dataset. Performance under operational distribution shift is not characterized.

Battery health prognostics has concentrated on SOH estimation and RUL prediction. Surveys by Che et al. [1] and Wang et al. [2] catalogue data-driven approaches. Representative methods include deep learning for SOH without degradation experiments [3], optimized XGBoost for capacity prediction [4], and cycle-based deep architectures [5]; Massaoudi et al. [6] review deep-learning prognostics. Hybrid and EIS-based RUL models [7]–[9], joint SOH/RUL prediction from single-cycle or discharge-fragment data [10], [11], and transfer learning for fast-charging profiles [12] round out the recent work. A common limitation is that validation is confined to clean laboratory splits of a single dataset. Performance under operational distribution shift is not characterized.

Related work B — merge four paragraphs into one (content identical, only paragraph breaks removed)	Probability calibration has a long lineage. Platt [13] introduced sigmoid scaling for support vector machines; Zadrozny and Elkan [14] extended calibration to tree ensembles; Niculescu-Mizil and Caruana [15] showed that [equation] ¶ ¶ boosted ensembles exhibit characteristic sigmoidal distortion. ¶ ¶ Naeini et al. [16] formalized ECE. Guo et al. [17] demonstrated that modern models are poorly calibrated in- ¶ ¶ distribution and that temperature scaling helps. Temperature scaling remains the default post-hoc method because it fits a single parameter, but that parametric form is too restrictive when the distortion is nonlinear, which is the regime produced by operational shift. Isotonic regression, used here, is the nonparametric counterpart, at the price of a fitted sample.	Probability calibration has a long lineage. Platt [13] introduced sigmoid scaling for support vector machines; Zadrozny and Elkan [14] extended calibration to tree ensembles; Niculescu-Mizil and Caruana [15] showed that boosted ensembles exhibit characteristic sigmoidal distortion. Naeini et al. [16] formalized ECE. Guo et al. [17] demonstrated that modern models are poorly calibrated in-distribution and that temperature scaling helps. Temperature scaling remains the default post-hoc method because it fits a single parameter, but that parametric form is too restrictive when the distortion is nonlinear, which is the regime produced by operational shift. Isotonic regression, used here, is the nonparametric counterpart, at the price of a fitted sample.
Related work C — heading — drop the letter numbering	C. Calibration under distribution shift	Calibration under distribution shift
Related work C — merge two paragraphs into one (only the last sentence changes)	Ovadia et al. [18] benchmarked predictive uncertainty under dataset shift and found calibration collapses before [equation] ¶ ¶ discrimination. Levi et al. [19] extended calibration evaluation to regression. Gawlikowski et al. [20] consolidated the uncertainty literature; Angelopoulos and Bates [21], Zhang and Wang [22], and Gibbs and Candès [23] provided conformal alternatives with distribution-free guarantees; Wu [24] surveyed emerging calibration and shift problems. This literature is dominated by image and NLP benchmarks. Batteries bring a specific constraint: the shift is structured (partial discharge truncates the voltage curve directly), and the deployment question is the cost of restoring calibration, not just its detection.	Ovadia et al. [18] benchmarked predictive uncertainty under dataset shift and found calibration collapses before discrimination. Levi et al. [19] extended calibration evaluation to regression. Gawlikowski et al. [20] consolidated the uncertainty literature; Angelopoulos and Bates [21], Zhang and Wang [22], and Gibbs and Candès [23] provided conformal alternatives with distribution-free guarantees; Wu [24] surveyed emerging calibration and shift problems. This literature is dominated by image and NLP benchmarks. Batteries bring a specific constraint: the shift is structured (partial discharge truncates the voltage curve directly), and the open deployment question is the cost of restoring calibration once it degrades.

Related work D — heading — drop the letter numbering	D. Domain randomization	Domain randomization
Related work E — heading — drop the letter numbering	E. Gap	Gap
Section III-A — one word	Two properties of the label drive the analysis below.	Two properties of the label shape the analysis below.
Section heading — drop the numbering	IV. EXPERIMENTAL DESIGN	Experimental Design
Section V-A (Clean baseline) — full paragraph	On the strictly held-out test partition, the frozen model's calibration is modest even on clean data: ECE ranges from 0.231 at H = 20 to 0.413 at H = 50 (Table III). These honest baselines make the degradation analysis below conservative, not inflated. Discrimination is modest throughout (AUC ≈ 0.48–0.59), typical of autocorrelated pre-failure labels; the practical headroom is in probability magnitudes, which is exactly the axis this paper targets. Clean ECE also varies with horizon, from 0.231 at H = 20 to 0.413 at H = 50, so horizon affects calibration even without any perturbation.	On the strictly held-out test partition, the frozen model's calibration is weak even on clean data: ECE ranges from 0.231 at H = 20 to 0.413 at H = 50 (Table III). Because that baseline is already weak, the degradation we report below is conservative. Discrimination is modest throughout (AUC ≈ 0.48–0.59), which is typical of autocorrelated pre-failure labels. The practical headroom sits in probability magnitudes, and horizon shifts calibration even without any perturbation.
**Section V-B (Calibration degrades) — full paragraph	Under perturbation, ECE rises at three of four horizons, from 0.231 to 0.305 at H = 20 and from 0.413 to 0.514 at H = 50, with little change at H = 10 (0.337 to 0.324; Table IV). The degradation saturates at the mildest severity: mean perturbed ECE is flat across severity	Under perturbation, ECE rises at three of four horizons, from 0.231 to 0.311 at H = 20 and from 0.413 to 0.507 at H = 50, with little change at H = 10 (0.337 to 0.328; Table IV). The degradation saturates at the mildest severity: mean perturbed ECE is flat across severity

, numbers revised**

levels, at 0.394, 0.382, 0.372, and 0.380 for S1 through S4 (Fig. 3). Fig. 4 explains the saturation. In full discharge Vmin reaches about 2.3 V; at the mildest truncation it jumps to about 3.2 V, a 37% shift the frozen calibrator cannot absorb. The reliability diagrams at H = 20 (Fig. 2) show the same pattern: every severity curve bends below the diagonal in the same probability region, and the curves cluster instead of spreading, confirming that saturation is a property of the shift rather than of severity. Within each severity the shift is dominated by the truncation draw rather than the noise level: mean ECE varies by at most 0.022 between S1 and S4 at any horizon, while the clean-to-perturbed gap reaches 0.101 at H = 50. Discrimination is stable by contrast: AUC moves by less than 0.1 at every horizon (0.594 → 0.609 at H = 20) and does not track the ECE collapse, so the perturbation attacks probability magnitudes, not ranking quality.

levels, at 0.395, 0.384, 0.372, and 0.382 for S1 through S4 (Fig. 3). Fig. 4 explains the saturation. In full discharge Vmin reaches about 2.3 V; at the mildest truncation it jumps to about 3.2 V, a 37% shift the frozen calibrator cannot absorb. The reliability diagrams at H = 20 (Fig. 2) show the same pattern: every severity curve bends below the diagonal in the same probability region, and the curves cluster instead of spreading, confirming that saturation is a property of the shift rather than of severity. Within each severity the shift is dominated by the truncation draw rather than the noise level: mean ECE varies by at most 0.047 between S1 and S4 at any horizon, while the clean-to-perturbed gap reaches 0.094 at H = 50. Discrimination is stable by contrast: AUC moves by less than 0.1 at every horizon (0.594 → 0.609 at H = 20) and does not track the ECE collapse. The perturbation shifts probability magnitudes while ranking quality stays intact.

Section V-C (Recalibration requirement) — full paragraph, numbers revised

Refitting the isotonic map on an independent field sample restores calibration, and the requirement is small. Mean held-out ECE falls monotonically with sample fraction, from 0.221 at 5% to 0.175 at 10%, 0.121 at 20%, and 0.080 at 50% (Fig. 5, Table V). The improvement saturates near a 20% fraction at every horizon. Paired bootstrap intervals for the 10% configuration exclude zero at all horizons, with gains from 0.125 [0.078, 0.168] at H = 10 to 0.389 [0.363, 0.413] at H = 50 and Cohen's dz from 1.2 to 6.7 (Table VI). For a deployment window of a few hundred cycles, a 20% sample is a modest collection effort, and it leaves the base model untouched. The curve is steep between 5% and 20% and flattens beyond: the marginal gain from 20% to 50% is only 0.041 in mean ECE, which bounds the operator's collection effort. The requirement is also approximately horizon-independent: at a 20% fraction, per-horizon means cluster between 0.098 (H = 10) and 0.149 (H = 30), within 0.05 of each other. Two checks support the sweep. Seed-to-seed spread of ECE at a

Refitting the isotonic map on an independent field sample restores calibration, and the requirement is small. Mean held-out ECE falls monotonically with sample fraction, from 0.221 at 5% to 0.175 at 10%, 0.121 at 20%, and 0.080 at 50% (Fig. 5, Table V). The improvement saturates near a 20% fraction at every horizon. Paired bootstrap intervals for the 10% configuration exclude zero at all horizons, with gains from 0.141 [0.113, 0.169] at H = 10 to 0.358 [0.333, 0.382] at H = 50 and Cohen's d_z from 1.8 to 6.2 (Table VI). For a deployment window of a few hundred cycles, a 20% sample is a small collection effort, and the base model stays untouched. The curve is steep between 5% and 20% and flattens beyond: the marginal gain from 20% to 50% is only 0.041 in mean ECE, which bounds the operator's collection effort. The requirement is also roughly the same at every horizon: at a 20% fraction, per-horizon means cluster between 0.098 (H = 10) and 0.149 (H = 30), within 0.05 of each other. Seed-to-

	10–20% fraction is 0.06–0.08, well below the gains of Table VI, so the requirement is not an artifact of any single augmentation. The paired rank-biserial correlation between recalibrated and uncalibrated ECE is 0.88 at H = 10 and 1.0 at longer horizons, so every condition improves and no outlier drives the mean.	seed spread of ECE at a 10–20% fraction is 0.06–0.08, well below the gains of Table VI, so the requirement is stable across individual augmentations. The paired rank-biserial correlation between recalibrated and uncalibrated ECE is 1.0 at every horizon: every condition improves, and no single condition drives the mean.
Section V-D (Domain randomization) — full paragraph, numbers revised	Retraining the base classifier on perturbed data is the standard alternative to post-hoc recalibration. A fresh XGBoost is trained on the clean training set augmented with perturbed copies at all four severities, for a total of 4,325 rows (the 865-row clean set plus four perturbed copies). Domain randomization wins ECE only at H = 10, at 0.074 against 0.199, and loses at every longer horizon, where recalibration reaches 0.125–0.146 while domain randomization stays between 0.314 and 0.381. The added complexity of generating perturbed training data and retraining is therefore not justified. Lightweight recalibration remains the preferred deployment strategy. The gap widens with horizon: at H = 50 domain randomization reaches 0.381 against 0.146 for recalibration.	Retraining the base classifier on perturbed data is the standard alternative to post-hoc recalibration. A fresh XGBoost is trained on the clean training set augmented with perturbed copies at all four severities, for a total of 4,325 rows (the 865-row clean set plus four perturbed copies). Domain randomization wins ECE only at H = 10, at 0.074 against 0.187, and loses at every longer horizon, where recalibration at a 10% sample reaches 0.149–0.195 while domain randomization stays between 0.314 and 0.381. The added complexity of generating perturbed training data and retraining is not justified. Lightweight recalibration remains the preferred deployment strategy. The gap widens with horizon: at H = 50 domain randomization reaches 0.381 against 0.149 for recalibration.
Section heading — drop the numbering	VI. Discussion	Discussion
Section VI-B (Deployment recipe) — full paragraph	The results support a concrete procedure: deploy the frozen model; collect about 20% of a field window from normal operation; extract the same seven per-cycle features; fit a fresh isotonic map on the raw probabilities versus observed outcomes; redeploy with the recalibrated map; refresh the calibrator periodically as conditions drift (Fig. 6). No base retraining, no original training data, and no new features are required. The recipe assumes the shift is static within a refresh window; under continuous drift the interval must shrink, and the monitoring	The results support a concrete procedure: deploy the frozen model, collect about 20% of a field window from normal operation, and refit a fresh isotonic map on the raw probabilities versus observed outcomes using the same seven per-cycle features (Fig. 6). Redeploy with the recalibrated map, and refresh the calibrator periodically as conditions drift. None of the steps touches the base model, the feature set, or the original training data. The recipe assumes the shift is static within a refresh window; under continuous drift the interval must

	loop of Section VI-A flags when it should.	shrink, and the monitoring loop of Section VI-A flags when it should.
Section VI-C (Limitations) — last sentence only	Within these limits, the answer to the motivating question is clear: calibration of a frozen battery hazard model degrades sharply under operational shift, and a field sample of roughly 20% of a deployment window restores it by refitting the calibrator.	None of these caveats changes the practical conclusion: calibration of a frozen battery hazard model degrades sharply under operational shift, and refitting the calibrator on roughly 20% of a deployment window restores it.
Section heading — drop the numbering	VII. Conclusion	Conclusion
Conclusion — full paragraph	This paper quantified the cost of restoring calibration for a battery hazard model under operational distribution shift. Three findings stand out. Calibration is fragile: ECE rises from 0.23–0.41 on clean data to 0.31–0.51 under perturbation and saturates at the mildest severity. The fix is cheap: a 20% field sample suffices to refit the isotonic map at every horizon, with gains well separated from zero. And the conventional fix, domain-randomized retraining, does not beat this lightweight maintenance step at longer horizons. Operators can keep a validated base model, collect a modest field sample, and maintain trustworthy probabilities through periodic recalibration. Future work targets non-Gaussian and coupled perturbation families that move degradation dynamics as well as observations, adaptive refresh schedules triggered by stream ECE, and replication across chemistries and model families.	This paper quantified the cost of restoring calibration for a frozen battery hazard model that must operate under operational distribution shift. Three findings stand out. Calibration is fragile: ECE rises from 0.23–0.41 on clean data to 0.31–0.51 under perturbation and saturates at the mildest severity. The damage is cheap to repair. A 20% field sample suffices to refit the isotonic map at every horizon, with gains well separated from zero. Domain-randomized retraining, the conventional alternative, never beats this lightweight maintenance step at longer horizons. Operators can keep their validated base model. Periodic recalibration on a modest field sample maintains trustworthy probabilities. Future work targets non-Gaussian and coupled perturbation families that move degradation dynamics as well as observations, adaptive refresh schedules triggered by stream ECE, and replication across chemistries and model families.
References heading — optional — keep it if you prefer the	VIII. References	(delete the line — the final version has no heading; references start directly and auto-number [1]–[29])

heading

Table IV — replace the ENTIRE table with this one

Delete your current Table IV and paste this one:

H	ECE clean	ECE perturbed
H=10	0.337	0.328
H=20	0.231	0.311
H=30	0.369	0.388
H=50	0.413	0.507

Table VI — replace the ENTIRE table with this one

Delete your current Table VI and paste this one:

<table>
<tr><th>H</th><th>ECE gain</th><th>95% CI | Cohen's d_z</th></tr>
<tr><td>H=10</td><td>0.141</td><td>[0.113, 0.169] | 2.18</td></tr>
<tr><td>H=20</td><td>0.14</td><td>[0.107, 0.176] | 1.76</td></tr>
</table>

2

H = 30 0.193 [0.162, 0.223] | 2.75

H = 50 0.358 [0.333, 0.382] | 6.21

In the header row, write the last column as **Cohen's d_z** with the z as a subscript.

Notes

- Entries marked *optional* are heading-numbering cosmetics (`C.` / `D.` / `E.` / `IV.` / `VI.` / `VII.`, leading space before `Results`); the final file uses unnumbered headings, but you can keep the numbers if the template wants them.
- In the merged-paragraph rows, `¶` marks where your document has a paragraph break; delete the break while replacing.
- `[equation]` marks an inline equation that stays untouched.
- Replace the exact old text (including numbers) — several rows carry revised statistics from the rerun.